AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 23 · STUDENT LESSON

Training a small image model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Show the pixels. Fit one scalar. Select with validation. Freeze before final release.

Trace a complete bounded image experiment with fixed features,one trained value and an auditable baseline.

Keep one habit: All six development images → complete model register → executable fit/select/baseline → pictured budget limits → gated final workflow → independent new model.

MAIN LESSON · SECTION 1

Which part of this image model actually learns?

> Trace fixed features, one trained scalar, validation selection and a teacher-gated final comparison.

Learning goals

I can trace a complete image model with fixed features and one trained weight.

I can select a **checkpoint** using a declared validation rule.

I can evaluate a frozen model against a **baseline** and limit the conclusion.

Retrieve the components and their limits

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why could pooling all four pixels of a 2 by 2 image hide whether the bright pixels are on the left or right?

> **My starting explanation**

End goal

By the end, I can explain **training** a small image model using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Trace a complete image model with fixed features and one trained weight

Select a **checkpoint** using a declared validation rule

Evaluate a frozen model against a **baseline** and limit the conclusion

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Start with complete images and a known arithmetic target

> The target is strictly greater left-column sum;equal sums belong to class0.

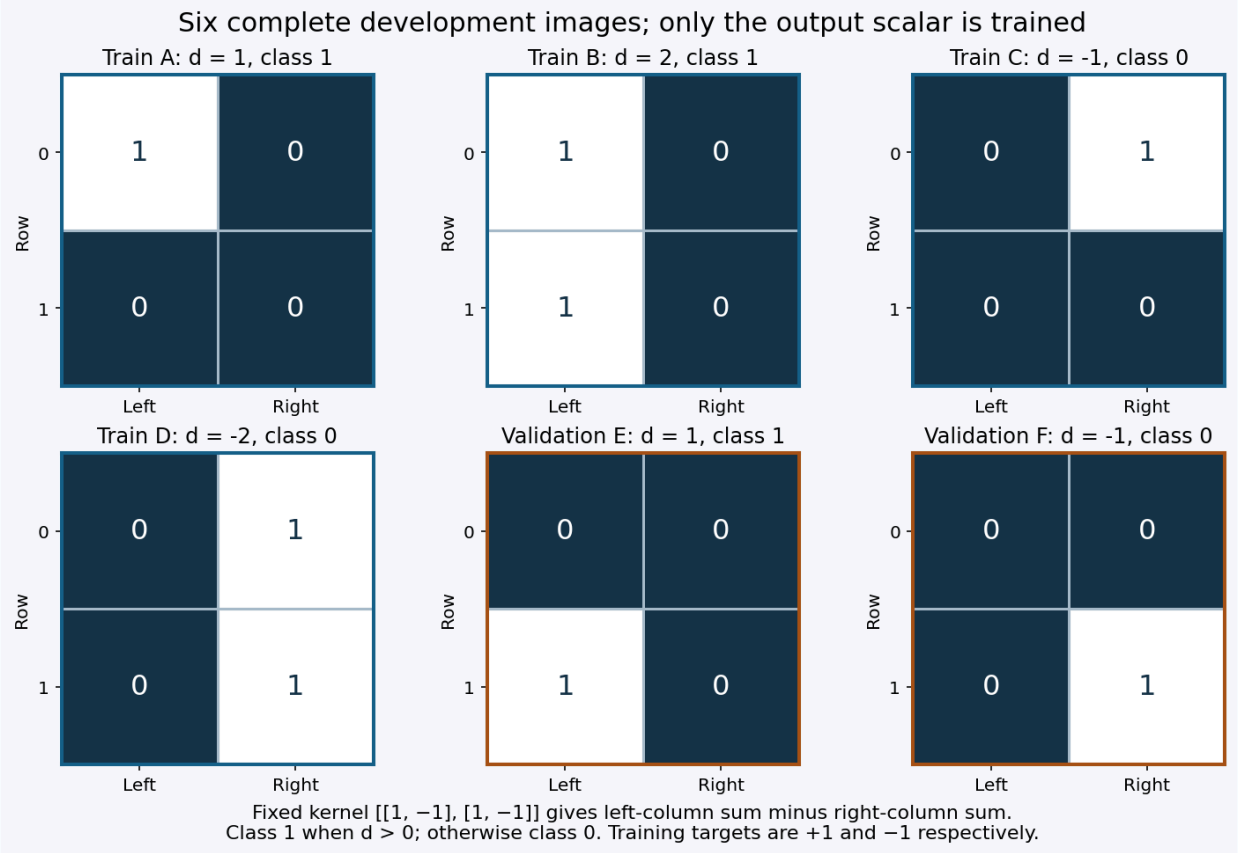
Target, complete development records and visible pixels

Use fictional 2 by 2 binary images, stored as two rows. Predict class 1 when the left column sum exceeds the right column sum; otherwise predict class 0. Equal sums belong to class 0.

This known arithmetic task makes every step checkable; it does not establish real-world image recognition.

Split and ID	Image rows	Class	Difference d
Train A	[1, 0] and [0, 0]	1	1
Train B	[1, 0] and [1, 0]	1	2

Split and ID	Image rows	Class	Difference d
Train C	[0, 1] and [0, 0]	0	-1
Train D	[0, 1] and [0, 1]	0	-2
Validation E	[0, 0] and [1, 0]	1	1
Validation F	[0, 0] and [0, 1]	0	-1



Fixed features, signed targets and strict decision rule

Apply the fixed kernel [[1, -1], [1, -1]] to the entire image to obtain d. Its negative gives -d. ReLU produces h1 = max(0, d) and h2 = max(0, -d). The **output** score is a(h1 - h2) = ad.

Only the shared scalar a is trained; both kernels stay frozen and there is no bias or pooling.

For squared-error **training**, map class 1 to target t = 1 and class 0 to t = -1. Predict class 1 only when score > 0; otherwise class 0. The score is not a probability.

Keep the teacher-held final cases unseen until the **checkpoint** is selected and recorded.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Run the complete restricted image-model experiment

> Separate the fixed image features from the one fitted parameter.

Complete frozen-versus-trained model register

Model component	Complete definition / training status
Input	One valid 2×2 binary image, rows then columns
Frozen kernels	K[[1, −1], [1, −1]]computes d; its negative computes−d
Frozen activation	h=[ReLU(d), ReLU(−d)], so h1−h2=d
Trainable output	Score a(h1−h2)=ad; coefficients[a, −a]are tied to one scalar, not two independent weights
Loss and targets	Ordinary MSE, class 1 target+1, class 0 target−1
Prediction	Class1 when score>0; otherwise 0, including equality
Training protocol	Four training rows, all per update, starta 0, rate.1, two updates; no bias or pooling

Model component	Complete definition / training status
Selection and baseline	Lowest observed validation MSE, earlier exact ties; training-majority class baseline chooses 0on a tie

Both gradients and all three saved checkpoints

Start a = 0. Use all four **training** examples in each update, ordinary mean squared error and learning rate 0.1. Save **checkpoint**s at updates 0, 1 and 2. The **training** loss is 2.5a squared - 3a + 1, with gradient 5a - 3.

Subtract learning rate times gradient; this gives a = 0.3 then a = 0.45.

Updates completed · a · Gradient for next update · Training MSE · Validation MSE · Training class correct/4 · Validation class correct/2

> **Updates completed:** 0
>
> **a:** 0
>
> **Gradient for next update:** -3
>
> **Training MSE:** 1
>
> **Validation MSE:** 1
>
> **Training class correct/4:** 2
>
> **Validation class correct/2:** 1

> **Updates completed:** 1
>
> **a:** 0.3
>
> **Gradient for next update:** -1.5
>
> **Training MSE:** 0.325
>
> **Validation MSE:** 0.49
>
> **Training class correct/4:** 4
>
> **Validation class correct/2:** 2

> **Updates completed:** 2

> **a:** 0.45
>
> **Gradient for next update:** Stop at declared budget
>
> **Training MSE:** 0.15625
>
> **Validation MSE:** 0.3025
>
> **Training class correct/4:** 4
>
> **Validation class correct/2:** 2

Protocol, selection and evidence boundary

The public program supplies all six A–Fdevelopment images, performs the two training updates, selects checkpoint 2(a≈.45)using validation MSE, and audits the frozen class 0 baseline. The two positive checkpoints have the same classification accuracy but different squared losses. Scores are not probabilities. Display rounding makes floating-point output readable; decisions use unrounded scores.

These constructed arrays and reused differences do not establish independent real-object coverage or broad image recognition. The three final cases and references remain in teacher materials until the actual freeze record is checked.

The code assumes valid 2 by 2 numeric images and nonempty labelled lists. It prints update, weight, **training** MSE and validation MSE. Validation examples are scored but never used in the gradient.

A single-channel filter spanning the whole image produces one value here, so no further spatial pooling is needed.

Complete runnable student development experiment

```
def feature(image):
    return image[0][0] + image[1][0] - image[0][1] - image[1][1]

train = [([[1, 0], [0, 0]], 1),
         ([[1, 0], [1, 0]], 1),
         ([[0, 1], [0, 0]], -1),
         ([[0, 1], [0, 1]], -1)]
valid = [([[0, 0], [1, 0]], 1),
         ([[0, 0], [0, 1]], -1)]
```

```
def mse(a, rows):
    return sum((a * feature(x) - t)**2 for x, t in rows) / len(rows)

a, checkpoints = 0.0, [0.0]
for _ in range(2):
    gradient = sum(2 * (a * feature(x) - t) * feature(x)
                   for x, t in train) / len(train)
    a -= 0.1 * gradient
    checkpoints.append(a)
for step, value in enumerate(checkpoints):
    print(step, round(value, 6), round(mse(value, train), 6),
          round(mse(value, valid), 6))
# Select using validation MSE only; exact ties keep the earlier checkpoint.
best_index = min(range(len(checkpoints)), key=lambda i: (mse(checkpoints[i],valid),i))
selected = checkpoints[best_index]
positive = sum(t == 1 for _,t in train)
negative = sum(t == -1 for _,t in train)
baseline_class = 1 if positive > negative else 0  # Declared class-0 tie rule.

def prediction(weight, image):
    d = feature(image)
    hidden = [max(0,d),max(0,-d)]
    score = weight*(hidden[0]-hidden[1])
    return d,hidden,score,int(score > 0)

def audit(weight, rows, ids, baseline):
    if len(rows) != len(ids) or not rows:
        raise ValueError('Need one ID per nonempty supplied case list')
    ledger = []
    for name,(image,target) in zip(ids,rows):
        d,h,score,pred = prediction(weight,image)
        ledger.append({'id':name,'d':d,'hidden':h,'score':score,
                       'class':int(target==1),'prediction':pred,'baseline':baseline})
    return ledger

print('selected checkpoint / weight', best_index,round(selected,6))
print('training class counts 1/0', positive,negative,'frozen baseline',baseline_class)
for split,rows,ids in [('training',train,['A','B','C','D']),('validation',valid,
['E','F'])]:
    ledger = audit(selected,rows,ids,baseline_class)
    for row in ledger:
        print(split,{**row,'score':round(row['score'],6)})  # Display rounding only.
    correct = sum(r['prediction']==r['class'] for r in ledger)
    base_correct = sum(r['baseline']==r['class'] for r in ledger)
    print(split,'candidate / baseline correct / total',correct,base_correct,len(ledger))
# This public boundary probe is not one of the teacher-held final cases.
empty_pattern = [[0,0],[0,0]]
print('public equal-column-sum probe
```

```
d/hidden/score/class',prediction(selected,empty_pattern))
print('Final cases absent. Save selected parameters and protocol before teacher release.')
# Fixed kernels, no bias/pooling; only a was fitted. Finite valid binary images assumed.
```

Connect the model diagram to each actual stage

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Reference: earlier compact training loop

```
def feature(image):
    return image[0][0] + image[1][0] - image[0][1] - image[1][1]

train = [([[1, 0], [0, 0]], 1),
         ([[1, 0], [1, 0]], 1),
         ([[0, 1], [0, 0]], -1),
         ([[0, 1], [0, 1]], -1)]
valid = [([[0, 0], [1, 0]], 1),
         ([[0, 0], [0, 1]], -1)]

def mse(a, rows):
    return sum((a * feature(x) - t)**2 for x, t in rows) / len(rows)

a, checkpoints = 0.0, [0.0]
for _ in range(2):
    gradient = sum(2 * (a * feature(x) - t) * feature(x)
                   for x, t in train) / len(train)
    a -= 0.1 * gradient
    checkpoints.append(a)
for step, value in enumerate(checkpoints):
    print(step, round(value, 6), round(mse(value, train), 6),
          round(mse(value, valid), 6))
```

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Keep training loss, validation selection and class accuracy distinct

> A later checkpoint is not selected merely because it exists.

Current two-update task

- Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the pictured doubled budget

- Change: Double the two-update budget while keeping the same **training** rule.

▼ **Check after predicting**

Prediction: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace each current-state gradient before the next update

> Keep the pictured single-pair experiment separate from the four-image training set.

Complete pictured configuration and separate scope

Train one tiny image score with a fixed budget and a separately reserved check.

Toy brightness difference d=left−right; score=a×d. **Training** pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate scalar brightness experiment:d2, target 2, starta 0, squared loss, rate 1/16. Two updates use gradients−8 then−4, giving a.5 then.75 and training losses 1 then.25. Its supplied reserved d1, target 1 pair has score.75, error.0625. Four updates instead reach.875 then.9375, training loss.015625.

This different target/rate does not provide the main image model’s final cases or a comparison of independent performance after choosing the budget.

Trace both updates and the supplied reserved calculation

AI ACADEMY / AI-5/23 · 02 · TRACE THE CASE

Follow the deciding steps

01 First update

Gradient 2(2a-2)×2=-8; new a = 0.5, training loss 1.

02 Second update

Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.

03 Validation score

At reserved d1, target 1: prediction 0.75, squared error 0.0625.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First update	Gradient 2(2a-2)×2=-8; new a = 0.5, training loss 1.
2. Second update	Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.
3. Validation score	At reserved d1, target 1: prediction 0.75, squared error 0.0625.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

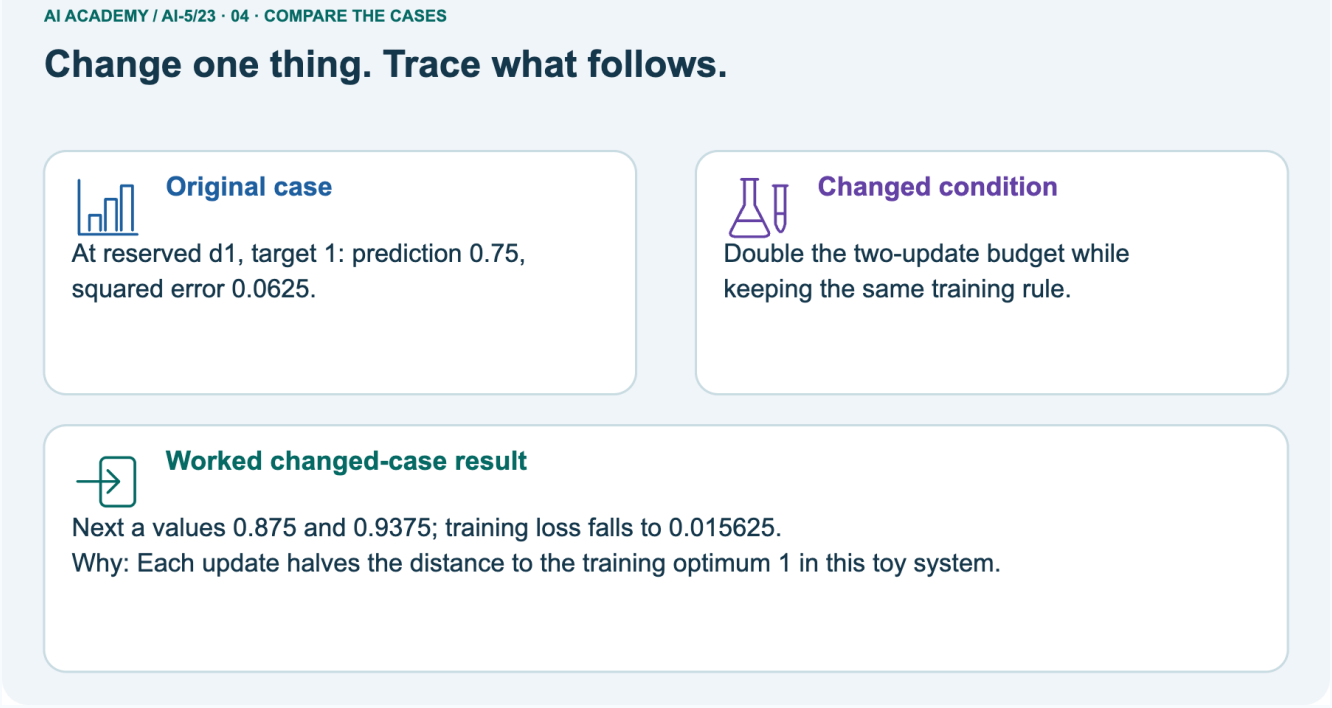
The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Predict the four-update training trace

- Change one thing: Double the two-update budget while keeping the same **training** rule. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain training progress and evaluation limits

- This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

- Read the labelled picture: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main E/Fscores at two saved checkpoints

Compare the saved checkpoints

UPDATE AND WEIGHT	TRAINING MSE	VALIDATION MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

At a = 0.3 and a = 0.45, compute the validation scores for E and F and their predicted classes. Why can validation accuracy stay the same while validation MSE changes?

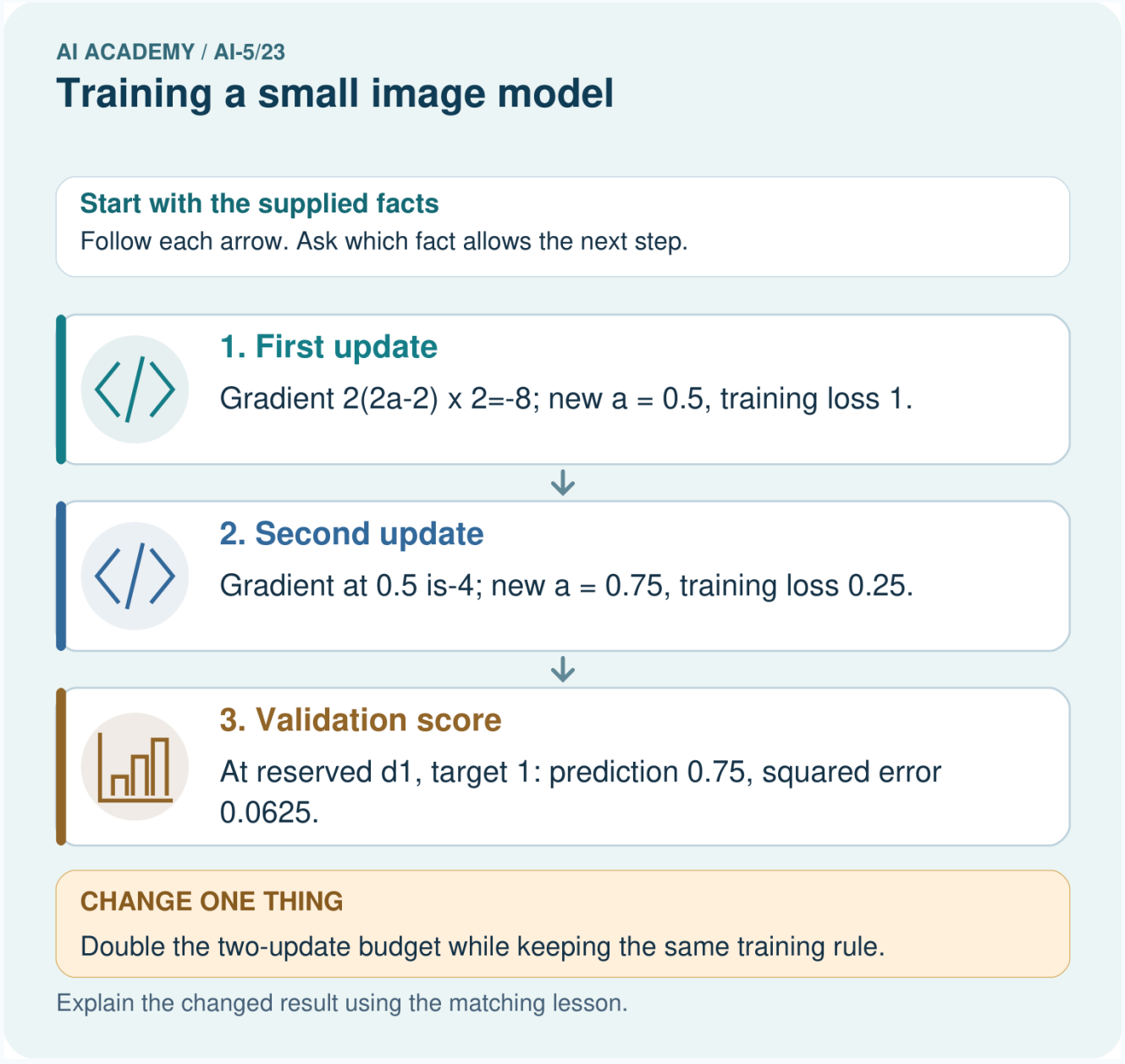
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training a small image model

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Freeze the selected system before opening final cases

> The teacher has the complete reserved reference fixture;student code does not include it.

Complete freeze record and baseline origin

Save a = 0.45, both fixed kernels, the **input** schema, target mapping, strict decision threshold, split IDs, learning rate, update budget and selection rule.

The **baseline** always predicts class 0: **training** classes are tied, and the declared tie rule chooses 0. It gets 2 of 4 **training** cases and 1 of 2 validation cases correct. The selected model gets all of those cases correct.

Before teacher final release	Required saved evidence
Selected system	Checkpoint2, a≈.45, exact saved numerical value and code/data version
Fixed computation	Both kernels, ReLU pair, tied output coefficients, no bias/pooling, input schema
Rules	Signed targets, strict score>0 class comparison, validation-MSE selection/earliest tie
Budgets and splits	A–Dtraining, E–Fvalidation, start 0, rate.1, two updates; record all scores/errors
Baseline	Training class 2/2 tie→always 0; training 2/4, validation 1/2
Actual gate	Teacher checks the saved record and records real status/date; no completed review is assumed here
After release	Record every final prediction, then compare with references and the same baseline; preserve failures and original frozen results

Teacher release and blank per-case scoring ledger

The teacher now supplies the three reserved final images. Record each score and **prediction** before comparing with the supplied labels.

Report correct count and denominator for both the frozen model and the same constant **baseline**. Do not tune the threshold or retrain after viewing final errors and still call that same set unseen.

Final row supplied by teacher after freeze · Score / predicted class before checking reference · Released reference · Baseline class · Candidate / baseline correct?

> **Final row supplied by teacher after freeze:** Release row 1
> **Score / predicted class before checking reference:** Your record
> **Released reference:** After prediction
> **Baseline class:** 0
> **Candidate / baseline correct?:** Your record

> **Final row supplied by teacher after freeze:** Release row 2
> **Score / predicted class before checking reference:** Your record
> **Released reference:** After prediction
> **Baseline class:** 0
> **Candidate / baseline correct?:** Your record

> **Final row supplied by teacher after freeze:** Release row 3
> **Score / predicted class before checking reference:** Your record
> **Released reference:** After prediction
> **Baseline class:** 0
> **Candidate / baseline correct?:** Your record

Bound the synthetic evidence and selection rule

- Because the images and target rule were constructed for this small exercise, even perfect results support only a narrow claim. Validation images differ from **training** pixels but reuse differences 1 and -1.

  This is a deliberately simple check of a procedure, not **evidence** of broad generalisation.

- Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the complete evidence chain with less help

> Show which data influenced fitting,which selected the checkpoint and which remain reserved.

Current task and response

- For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.

At a = 0.3 and a = 0.45, compute the validation scores for E and F and their predicted classes. Why can validation accuracy stay the same while validation MSE changes?

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example "**Training** a small image model". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Training** a small image model. Proposed explanation: "All parameters in a trained model must have been learned". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured budget reasoning

- Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the independent new scalar-model case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Check both updates

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run or trace a complete development-to-freeze investigation

> Every public development input and the later teacher-release workflow is explicit.

Current update task

Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Complete development probes and honest missing-final status

Complete development-model probe	Expected answer / action
Normal A–Fdevelopment run	Weights0/.3/.45; validation MSE 1/.49/.3025; select update 2
Baseline training-class tie	Always0 chosen from training counts; training 2/4, validation 1/2
Public equal-column-sum image[[0, 0], [0, 0]]	d0, h[0, 0], score 0, class 0; under signed target–1, squared error 1 despite correct class
Changing a.3→.45on validation E/F	Class predictions stay 1/0; MSEfalls.49→.3025
Missing row/pixel or nonbinary image	Outside valid 2×2 binary schema; review before fitting or prediction
A row ID or split label appended as a feature	Violates the declared input; IDs audit membership, not image computation
Final data not yet teacher-released	Record not evaluated; do not copy worked development scores as final evidence
Adjust threshold after final errors	New development revision; preserve the original frozen report and obtain fresh final evidence
Call the fixed kernels learned image features	Unsupported:only the tied output scalar was fitted

Predict,fit,select,audit and save the actual freeze record

1. Predict every checkpoint and all A–Fhidden vectors, scores and classes before running or tracing the complete student program in section 3. Retain the actual output and selected parameter, then complete the freeze checklist. The original compact loop below remains a shorter training-only reference; it does not itself select a model or audit the baseline.

   The teacher releases final cases only after checking the actual saved record. Score all released rows without retuning, and report count/denominator, baseline comparison, signed-loss definition and limits. This is a reproducible arithmetic image experiment, not a trained full vision network.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Reference: original training-only loop

```
def feature(image):
    return image[0][0] + image[1][0] - image[0][1] - image[1][1]
```

```
train = [([[1, 0], [0, 0]], 1),
         ([[1, 0], [1, 0]], 1),
         ([[0, 1], [0, 0]], -1),
         ([[0, 1], [0, 1]], -1)]
valid = [([[0, 0], [1, 0]], 1),
         ([[0, 0], [0, 1]], -1)]

def mse(a, rows):
    return sum((a * feature(x) - t)**2 for x, t in rows) / len(rows)

a, checkpoints = 0.0, [0.0]
for _ in range(2):
    gradient = sum(2 * (a * feature(x) - t) * feature(x)
                   for x, t in train) / len(train)
    a -= 0.1 * gradient
    checkpoints.append(a)
for step, value in enumerate(checkpoints):
    print(step, round(value, 6), round(mse(value, train), 6),
          round(mse(value, valid), 6))
```

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: choose a checkpoint for a different signed-target model

> Use its new training/validation differences,gradient,rate and budget.

Complete independent scalar-model comparison

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Choose a checkpoint independently

Attempt this without hints

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep fixed features and learned values in separate project fields

> The project report should identify the exact frozen system and its real evidence.

Project artifact and evidence table

Add a labelled **training** a small image model **evidence** sheet to Image-pattern Network Investigation. Show how you can trace a complete image model with fixed features and one trained weight.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one selected checkpoint and one bounded claim

> Name what learned,what stayed fixed and which evaluation remains unperformed.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Trace a complete image model with fixed features and one trained weight

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Frozen feature	A feature computation whose parameters are kept unchanged during training
Checkpoint	Saved model parameters at a particular training stage
Selection metric	The quantity used under a declared rule to choose a model
Baseline	A simple comparison method evaluated on the same cases

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.
- Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?
- After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent signed-target checkpoint-selection case for the first individual attempt.

Current explanation and response

- **Training** adjusted one weight in a model with hand-chosen features. It did not discover the left-versus-right kernel.

  The arithmetic rule itself would solve the declared task exactly; the trained model helps us study **fitting** and evaluation, not demonstrate that learning is necessary here.

- Report the fixed and trained components separately, selected **checkpoint**, per-case final **prediction**s, **baseline** results and limitations.

  If a final case fails, preserve that failure and design a new experiment with new reserved evaluation data. Next week examines augmentation and leakage between related images.

- After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

> **A teammate claims the model learned its own image features and that choosing the best score on the final test would improve the experiment. Correct both claims and name the evidence needed in the final report.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Compare the saved checkpoints

UPDATE AND WEIGHT	TRAINING MSE	VALIDATION MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Train one tiny image score with a fixed budget and a separately reserved check.

By the end, I can	How I will show it
Trace a complete image model with fixed features and one trained weight	Explain the deciding step, show a trace or test, and state one limit.
Select a checkpoint using a declared validation rule	Explain the deciding step, show a trace or test, and state one limit.
Evaluate a frozen model against a baseline and limit the conclusion	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Train one tiny image score with a fixed budget and a separately reserved check.
- Which labelled step matches this detail? At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.
- 1. Validation score
- 2. First update
- 3. Second update
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Validation score At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Changed-case question and optional hint

- Changed-case practice: Double the two-update budget while keeping the same **training** rule. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

Repair and retain the record

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

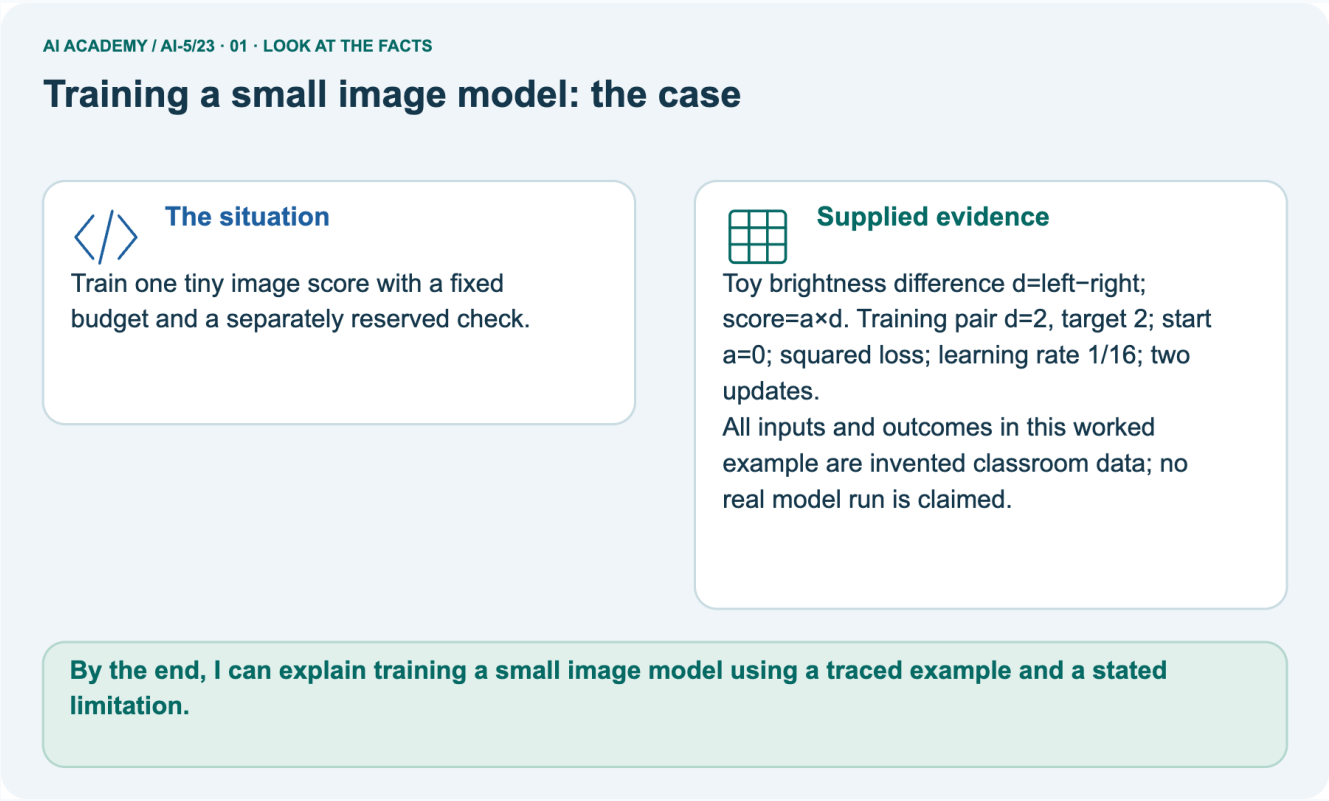
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Training a small image model: the case



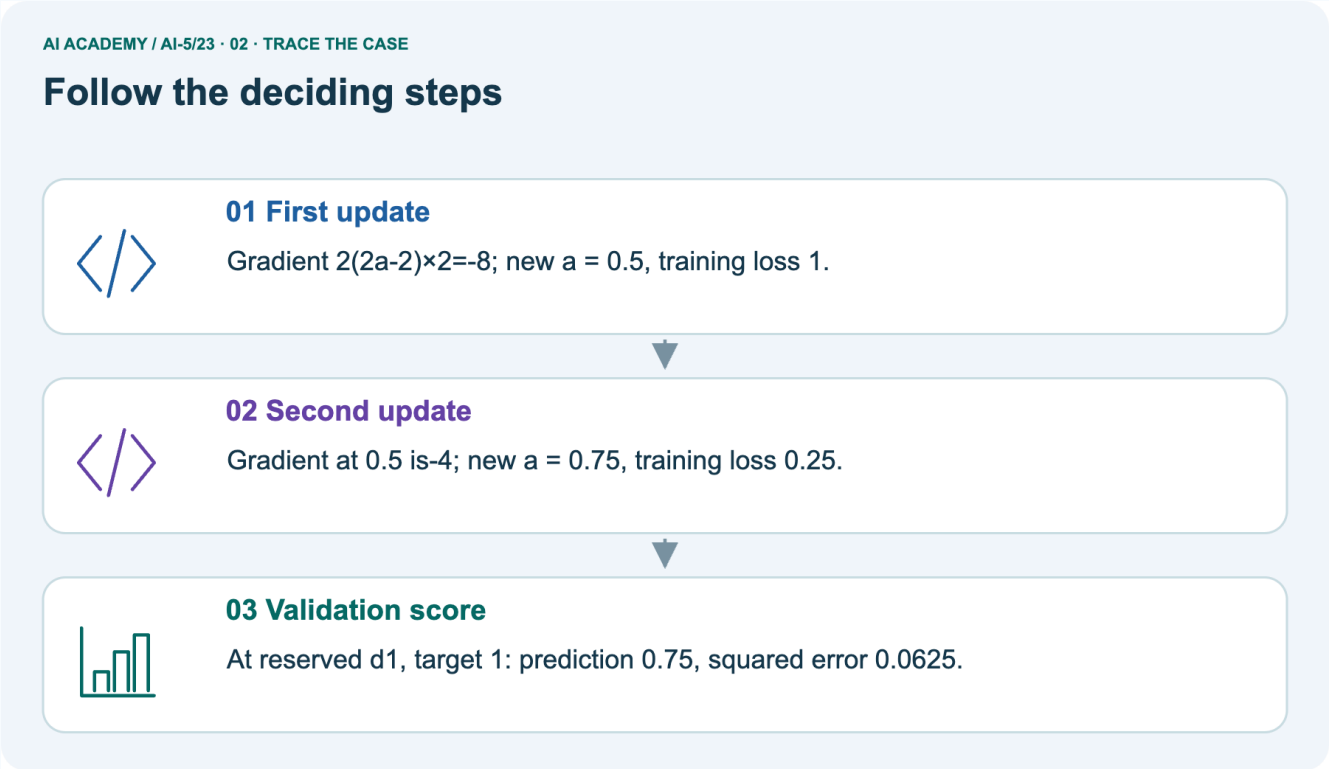
Follow the labelled picture with the matching explanation. Open the vector diagram

Train one tiny image score with a fixed budget and a separately reserved check. Supplied facts: Toy brightness difference d=left−right; score=a×d.

Training pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First update: Gradient 2(2a-2)×2=-8; new a = 0.5, **training** loss 1. → Second update: Gradient at 0.5 is-4; new a = 0.75, **training** loss 0.25. → Validation score: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

Point to each numbered stage and explain which recorded **input** it uses.

03. Compare the saved checkpoints

AI ACADEMY / AI-5/23 · 03 · INSPECT THE LESSON MODEL

Compare the saved checkpoints

Update and weight	Training MSE	Validation MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

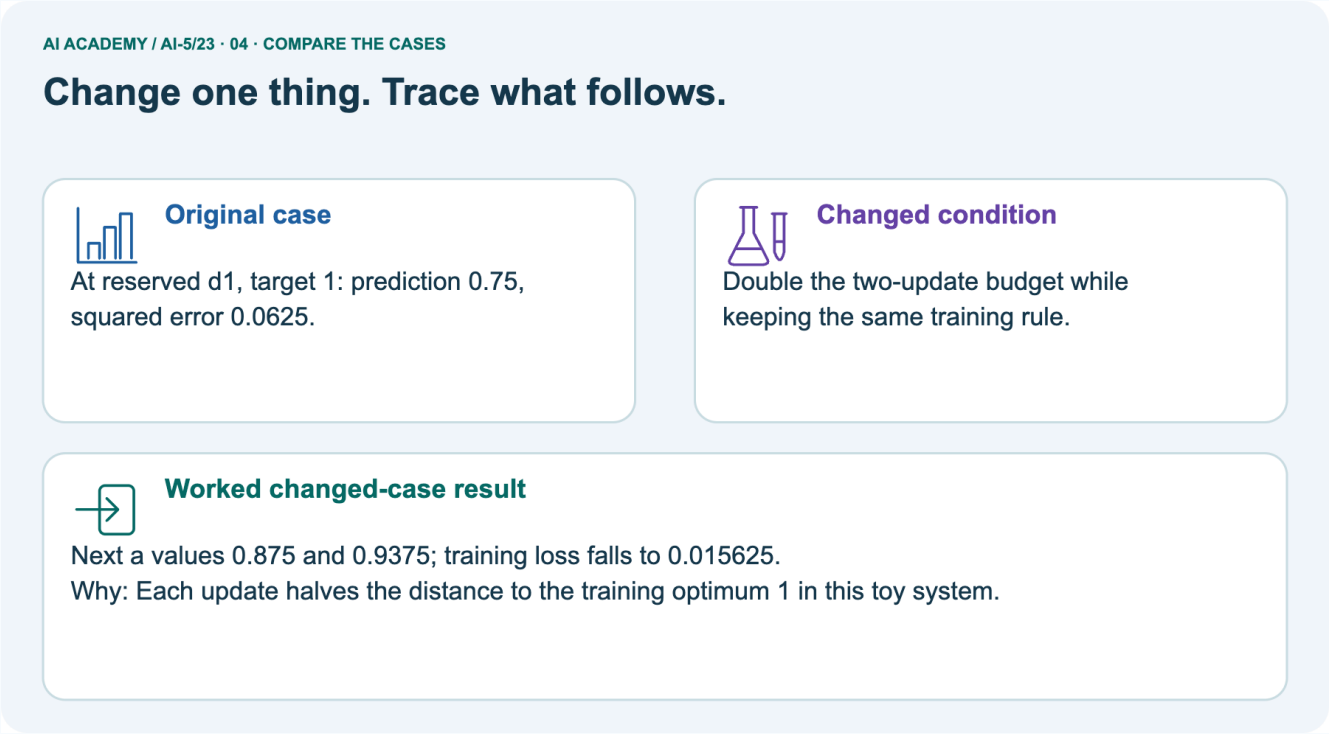
Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the saved **checkpoint**s. 0 with a 0 | 1 | 1. 1 with a 0.3 | 0.325 | 0.49. 2 with a 0.45 | 0.15625 | 0.3025

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



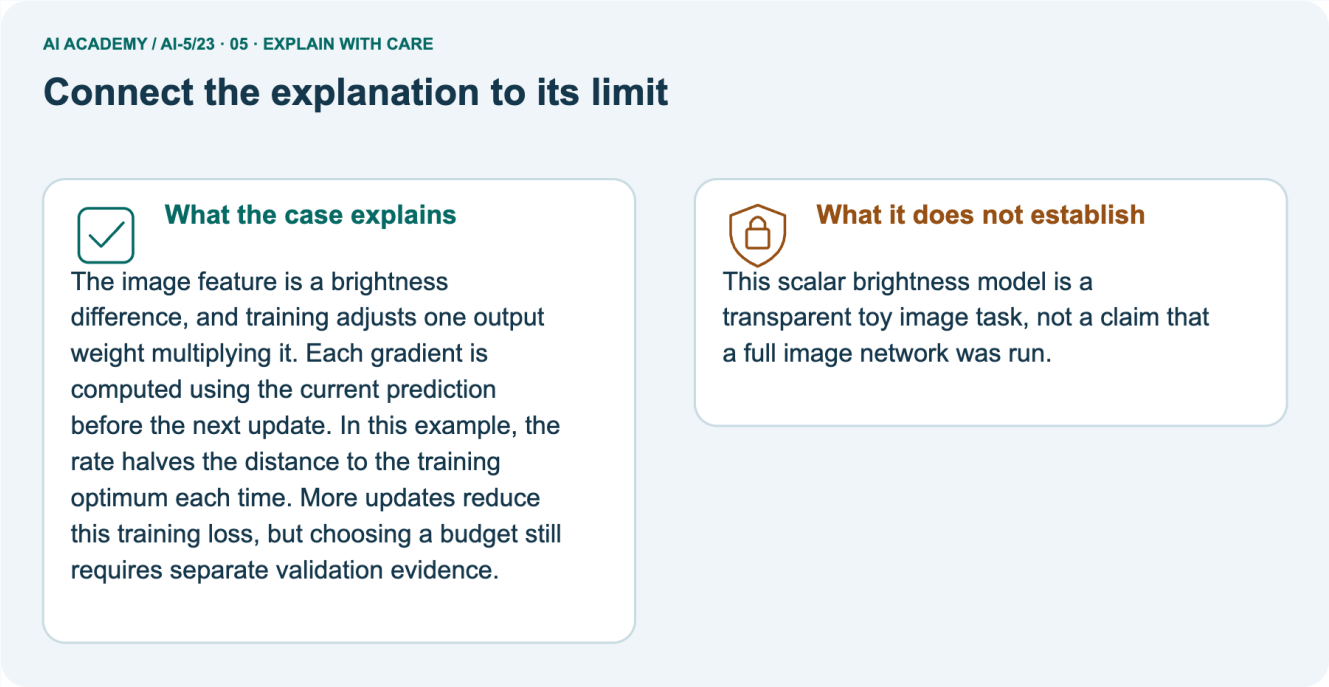
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625. Changed condition: Double the two-update budget while keeping the same **training** rule.

Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

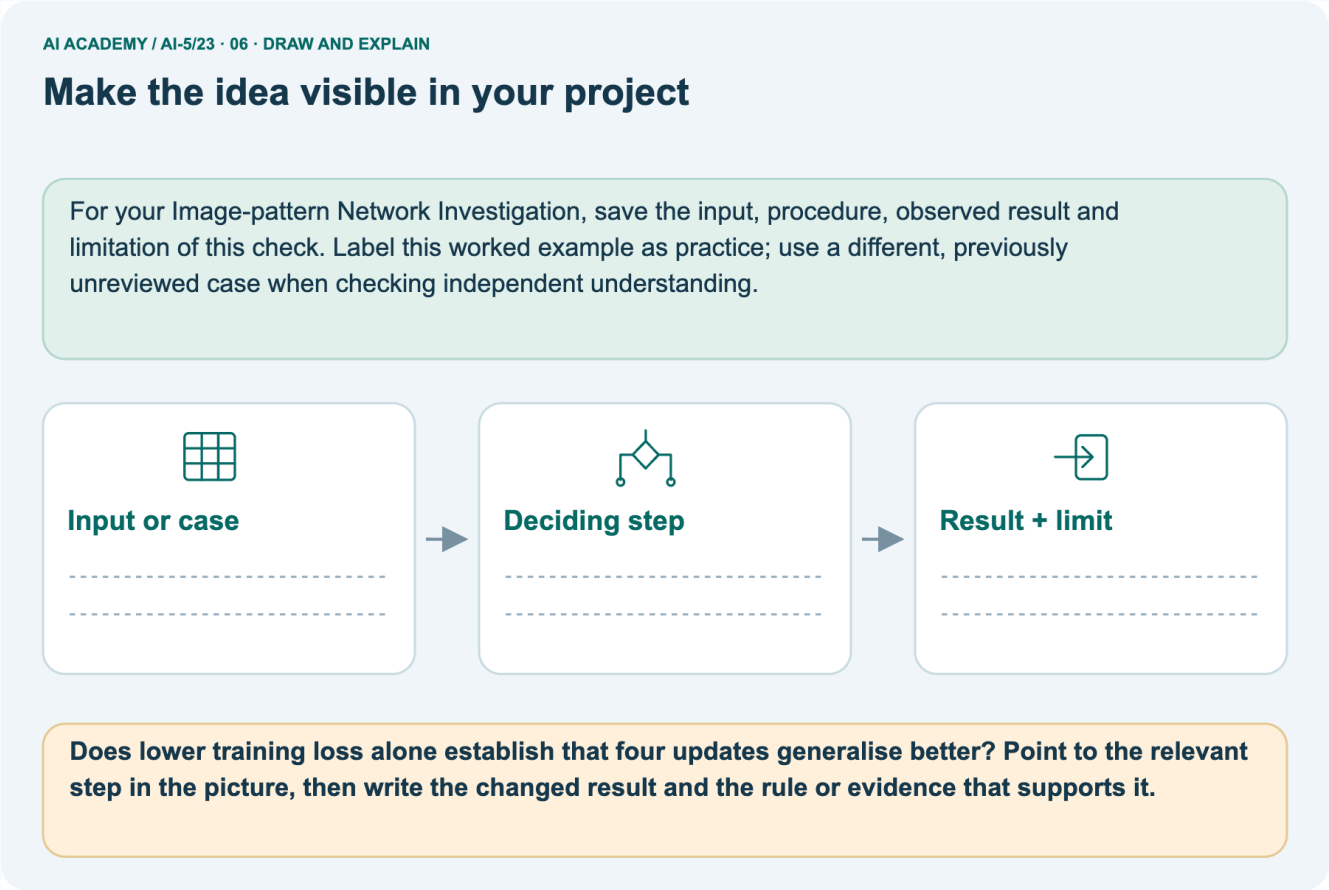
The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Limit: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Training a small image model: the case

02. Follow the deciding steps

03. Compare the saved checkpoints

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.